Moving Average Signal Analysis

Portfolio Builder

1 Objective

The objective of this test is to examine whether the position of a stock's current price relative to its recent moving average contains predictive information about its future returns.

The analysis uses a 20-day moving average and a 20-day forward return. The factor is evaluated using cross-sectional decile analysis, Information Coefficient (IC), a long-short portfolio, and a Heteroskedasticity and Autocorrelation Consistent (HAC) test.

2 Data

The dataset contains OHLCV data for 24 stocks.

- Valid observations: 34,656
- Number of stocks: 24
- Date range: 5 October 2016 to 5 August 2022

3 Signal Construction

The previous 20-day moving average is calculated as

$$MA_{20,t} = \frac{1}{20} \sum_{k=1}^{20} Close_{t-k}.$$

The Moving Average Signal is defined as

$$MA\ Signal_t = \frac{Close_t}{MA_{20,t}} - 1.$$

A positive value indicates that the current price is above the previous 20-day moving average, while a negative value indicates that the price is below it.

The future 20-day return is defined as

$$R_{t,t+20} = \frac{Close_{t+20}}{Close_t} - 1.$$

4 Signal Statistics

The distribution of the Moving Average Signal was:

The average signal was approximately 0.72%, indicating that stocks were, on average, slightly above their previous 20-day moving average.

Statistic	Value
Mean	0.007152
Standard deviation	0.051758
Minimum	-0.503455
25th percentile	-0.021135
Median	0.006675
75th percentile	0.036253
Maximum	0.311076

Table 1: Moving Average Signal statistics

5 Decile Analysis

Stocks were sorted into ten cross-sectional groups each day according to their Moving Average Signal.

Group 1 represents stocks with the lowest signal values, while Group 10 represents stocks with the highest signal values.

Group	Average Future 20-Day Return
1	1.8940%
2	1.9392%
3	2.1428%
4	2.1707%
5	2.0568%
6	1.3983%
7	1.6681%
8	1.4852%
9	1.0113%
10	1.0197%

Table 2: Future returns by Moving Average Signal decile

The lowest-signal group produced an average future return of 1.8940%, while the highest-signal group produced 1.0197%.

Therefore, the low-minus-high spread was

$$1.8940\% - 1.0197\% = \boxed{0.8743\%}.$$

The general pattern indicates that stocks with higher Moving Average Signals tended to have lower subsequent returns. This is more consistent with a reversal relationship than with conventional trend-following behaviour.

6 Information Coefficient

The daily cross-sectional Spearman Information Coefficient between the Moving Average Signal and the future 20-day return was calculated.

The mean IC of -0.057942 indicates a negative relationship between the Moving Average Signal and subsequent returns.

The low positive-IC fraction of 42.24% also indicates that the relationship was more frequently negative than positive.

Statistic	Value
Mean IC	-0.057942
Median IC	-0.049565
IC standard deviation	0.248958
Positive IC fraction	42.24%
t-statistic	-8.843988
p-value	< 0.001

Table 3: Information Coefficient results

The ordinary statistical test gives a very small p-value. However, the overlapping nature of 20-day forward returns must be considered when evaluating portfolio-level significance.

7 Long-Short Portfolio

A long-short portfolio was constructed by taking:

- Long position: lowest Moving Average Signal decile
- Short position: highest Moving Average Signal decile

The resulting spread was

$$R_{LS} = R_{G1} - R_{G10}.$$

The portfolio statistics were:

Statistic	Value
Mean return	0.008743
Median return	0.006517
Standard deviation	0.071228
Annualized return	220.3321%
Annualized volatility	113.0714%
Sharpe-like ratio	1.9486
Ordinary t-statistic	4.664536
Ordinary p-value	0.000003

Table 4: Moving Average Signal long-short portfolio

The long-short portfolio produced a positive mean spread of 0.8743%. The ordinary t-test indicates statistical significance.

The annualized return should not be interpreted as an expected annual portfolio return because the strategy uses overlapping 20-day forward returns.

8 HAC / Newey-West Test

Since the 20-day forward returns overlap across consecutive observations, the returns can exhibit autocorrelation. Therefore, a HAC/Newey-West standard error with 20 lags was used.

The HAC p-value is 0.065613. Therefore, the long-short portfolio does not satisfy the conventional 5% significance threshold after accounting for autocorrelation and overlapping returns.

This makes the HAC result more conservative than the ordinary t-test.

Statistic	Value
Mean return	0.008743
HAC standard error	0.004749
HAC t-statistic	1.841057
HAC p-value	0.065613

Table 5: HAC/Newey-West results

9 Interpretation

The Moving Average Signal shows a clear negative relationship with subsequent 20-day returns.

The decile analysis shows that stocks with higher Moving Average Signals generally produced lower future returns. The mean IC of -0.057942 provides additional evidence of this negative relationship.

Thus, the signal behaves more like a short/medium-term reversal indicator than a conventional trend-following indicator.

However, the HAC/Newey-West test gives a p-value of 0.065613, which is above the 5% significance level. Therefore, the evidence is not strong enough to classify the signal as a statistically significant standalone factor after accounting for overlapping returns.

10 Conclusion

The Moving Average Signal does not provide strong evidence for conventional moving-average momentum in this dataset.

Instead, it shows a reversal-like relationship:

$$\boxed{\text{Higher Moving Average Signal} \longrightarrow \text{Lower Subsequent Return}}$$

The signal has:

- Negative mean IC: -0.057942
- Positive low-minus-high spread: 0.8743%
- Significant ordinary long-short test
- HAC p-value of 0.065613

Therefore, the Moving Average Signal is classified as a **supporting / potential reversal factor**, rather than a primary standalone alpha factor.

It may be useful in combination with the stronger reversal signal or other factors, but its standalone statistical evidence should be treated cautiously.